INTERRUPTION SCIENCE · A GYR-O SITE

Interruption Science · what it costs to be pulled away, and how people get back

WHAT THIS IS

Being interrupted is ordinary; getting back is the expensive part. Researchers have spent two decades timing it: shadowing office workers, running laboratory tasks, and modeling memory.

Here are their findings in plain words, read carefully, with every source previewed in the middle column before you decide whether to leave this page for it.

first published 3 November 2026 · nine sources · no trackers, nothing stored

THE PAGE LAYOUT BELOW IS STANDARD ON EVERY GYR-O SITE

A PLAIN ANSWERS ✓

– What is interruption science?

The study of what happens when work is interrupted: what it costs, how people get back to what they were doing, and how tools could make that easier. It sits mainly in human–computer interaction, the field that studies how people use computers, with roots in the psychology of memory and attention.

It works two ways. In the field, researchers sit beside people at real jobs and time every switch[1]. In the laboratory, they interrupt people on purpose, at chosen moments, and measure how long it takes to resume[4][6].

– Does it really take 23 minutes to get back on track?

Not in the way it is usually quoted. The figure comes from Gloria Mark at the University of California, Irvine, who shadowed office workers and timed their days[1]. She found that an interrupted piece of work was usually picked up again the same day, about 82% of the time, and that on average it was resumed **23 minutes and 15 seconds** later[2].

That is the time until people went *back to* the interrupted work, not the time it took them to refocus once they were back. In between, they typically worked on about two other things first. The cost she points to is the effort of switching back from those.

– How often does it happen?

Very often. In Mark's study people spent about three minutes on a single event, such as a phone call or a document, before switching to something else, and about ten and a half minutes on a larger project, which she calls a *working sphere*, before switching to another one. They juggled about twelve working spheres a day[2].

– Do people interrupt themselves?

Yes, nearly as often as others interrupt them. In Mark's study about 44% of interruptions were self-started: nobody came to the desk or rang the phone; people simply switched[2].

– Are interruptions always harmful?

Not always, and the research is honest about the uncertainty. In the field, interruptions about the same work were described as helpful, and Mark's informants called them *interaction*, not distraction[2].

But when Mark and colleagues tested this in the laboratory, the subject of the interruption made no difference. What they found instead was surprising: interrupted people finished *faster*, with no loss of quality, and paid for it in stress, frustration, time pressure and effort[3]. The cost moved from the clock to the person.

– What happens in the moment you are pulled away?

Two short gaps get measured. The **interruption lag** is the moment between noticing an interruption (the phone starts ringing) and turning to it. The **resumption lag** is the time, after it is over, before you are working on the original task again.

A small warning helps. People who had eight seconds of notice before an interruption used it to prepare, and resumed the original task more quickly than people interrupted without notice[4]. The picture at the top of this page shows both lags.

– Why does the last task linger in your head?

Sophie Leroy called it **attention residue**: when you switch away from a task you have not finished, part of your attention stays with it, and you start the next one with less[5]. It is why a switch can cost more than the seconds it takes.

– Does the timing of an interruption matter?

Yes. Experiments that interrupted people at different moments within the same task found that the moment mattered: interruptions at natural break points, between steps, were easier on people's emotional state and on how they felt about whoever interrupted them[6]. The practical lesson for software is to hold a notification for the next break, not to deliver it mid-step.

– Why do we keep leaving for things that turn out not to be worth it?

Because we choose where to go from very little evidence. **Information foraging** theory, from researchers at Xerox PARC, describes people following an *information scent* the way an animal follows a trail, judging each link by the few words around it[7]. A weak scent sends people on trips that a short preview would have saved. That is why every source on this page is previewed before its link.

– Do even tiny switches cost anything?

A little, many times over. A classic model of skilled computer use puts the time just to move a hand between keyboard and mouse at about 0.4 seconds each way, before any pointing at all[8]. It is nothing once and something at several hundred times a day.

– Where can I read the research myself?

Many of the papers cited here are free to read in full, and a long-running online bibliography of interruption research collects a great many more[9]. For a book-length account for general readers, Gloria Mark's *Attention Span* (2023) brings her two decades of measurement together.

– Who made this site, and why?

It grew out of **Round Trip Distraction**, a user-experience experiment by Walter Wilkinson of Worland, Wyoming, that catalogs the moments when people leave their work and have to find their way back, and builds small tools to measure them. This site is the reading list behind that work: the findings it rests on, stated carefully.

B SOURCES ✓

HOW TO READ A CARD

Every card below reads the same way, top to bottom. Parts 4 and 5 are the ones a bare footnote never gives you.

1. **Number and kind** — what sort of thing it is
2. **Title** — what it is called
3. **Provenance** — who made it, when, under what license
4. **Preview** — what you would find if you went there
5. **Why it is cited** — what it is doing on this page
6. **The link** — last, and only last

INTERNAL LINKS

The two most recent content changes, and the record behind them.

3 November 2026 — first publish: twelve plain answers, in column A

From what interruption science is to where to read it yourself, with the 23-minute figure read as the study actually reported it.

3 November 2026 — the two lags, drawn: in column C

Interruption lag and resumption lag on one timeline, the picture at the top of the page, explained step by step.

Chronicle — what has been happening, and why

The whole record in plain language, newest first: what changed, what it was for, and what is still unresolved.

EXTERNAL LINKS

Every reference to some other website, previewed. The link is always the last line — read the card first and decide whether you need the round-trip distraction of visiting the original at all.

1 STUDY · FIELD OBSERVATION

No Task Left Behind? Examining the Nature of Fragmented Work

Gloria Mark, Victor M. González and Justin Harris · University of California, Irvine · CHI 2005 conference, Portland, Oregon · pages 321–330

Researchers shadowed information workers through their working days and timed every switch between activities and between larger projects, which the paper calls working spheres. It describes how fragmented the day is, how often work is interrupted, and how and when interrupted work is resumed. Ten pages, free to read as a PDF.

Why it is cited: as the original study behind almost every figure in column A, including the much-quoted 23 minutes.

→ interruptions.net/literature/Mark-CHI05-p321-mark.pdf

2 INTERVIEW · PLAIN LANGUAGE

Too Many Interruptions at Work?

Gallup Business Journal · interview with Gloria Mark by Jennifer Robison · 8 June 2006

Mark explains her study in conversation: the 82% of interrupted work resumed the same day, the average of 23 minutes and 15 seconds before it was, the two intervening tasks, the 44% of interruptions that were self-started, and the difference between helpful interaction and harmful distraction.

Why it is cited: because it states the figures in plain words, exactly as she put them, which is the best guard against the usual misquotation.

→ news.gallup.com/businessjournal/23146/too-many-interruptions-work.aspx

3 STUDY · LABORATORY EXPERIMENT

The Cost of Interrupted Work: More Speed and Stress

Gloria Mark, Daniela Gudith and Ulrich Klocke · UC Irvine and Humboldt University Berlin · CHI 2008 conference, Florence

A laboratory test of whether interruptions about the same work hurt less. They did not: the subject of the interruption made no difference. Interrupted people finished their tasks faster with no loss of quality, and reported more stress, frustration, time pressure and effort. Free to read as a PDF.

Why it is cited: as the honest counterweight to the idea that interruptions simply waste time: here the time was made up, and the person paid instead.

→ interruptions.net/literature/Mark-CHI08.pdf

4 STUDY · LABORATORY EXPERIMENT

Preparing to resume an interrupted task: effects of prospective goal encoding and retrospective rehearsal

J. Gregory Trafton, Erik M. Altmann, Derek P. Brock and Farilee E. Mintz · Naval Research Laboratory and Michigan State University · International Journal of Human-Computer Studies 58 (2003), pages 583–603

Defines the interruption lag, the gap between an alert and the interruption proper, and shows that people given eight seconds of warning prepared and resumed the original task faster than people interrupted at once; those interrupted at once improved with practice. Free to read as a PDF.

Why it is cited: for the two lags drawn at the top of the page, and for the finding that a few seconds of warning help.

→ interruptions.net/literature/Trafton-IJHCS03.pdf

5 STUDY · ORGANIZATIONAL PSYCHOLOGY

Why is it so hard to do my work? The challenge of attention residue when switching between work tasks

Sophie Leroy · Organizational Behavior and Human Decision Processes 109, no. 2 (2009), pages 168–181

Introduces attention residue: after switching away from an unfinished task, people keep thinking about it, and perform worse on the task they switched to. The record page lists the article and its abstract; the full text is in the journal.

Why it is cited: for the name and the idea behind “why the last task lingers.”

→ econpapers.repec.org/RePEc:eee:jobhdp:v:109:y:2009:i:2:p:168-181

6 STUDY · LABORATORY EXPERIMENT

If Not Now, When? The Effects of Interruption at Different Moments Within Task Execution

Piotr D. Adamczyk and Brian P. Bailey · University of Illinois at Urbana-Champaign · CHI 2004 conference, Vienna

Interrupts people at different points inside the same tasks, chosen from models of where the natural breaks fall, and measures performance, emotional state and how people felt about the interrupter. Different moments had different effects, which points to software that holds an interruption for a better moment. Free to read as a PDF.

Why it is cited: for the finding that the timing of an interruption matters, and the case for holding notifications until a break.

→ hci.rwth-aachen.de/materials/conferences/CHI2004/1p271.pdf

7 REFERENCE · THEORY

Information foraging

Wikipedia · on the theory of Peter Pirolli and Stuart Card, Xerox PARC, published in Psychological Review 106 (1999)

An encyclopedia account of information foraging theory: people seeking information behave like foragers, follow an information scent from the cues around each link, and leave a source when the expected gain drops. Includes the central terms and the original references.

Why it is cited: for "information scent" and the reason this page previews every source before its link.

→ en.wikipedia.org/wiki/Information_foraging

8 REFERENCE · MODEL

Keystroke-level model

Wikipedia · on Stuart K. Card, Thomas P. Moran and Allen Newell, Communications of the ACM 23, no. 7 (1980)

A summary of the keystroke-level model, which predicts how long a skilled user takes to do a routine task by adding up standard times for its small operations: pressing a key, pointing, moving the hand between keyboard and mouse, thinking. It lists the standard values, including about 0.4 seconds for that hand movement.

Why it is cited: for the 0.4-second figure in "Do even tiny switches cost anything?"

→ en.wikipedia.org/wiki/Keystroke-level_model

9 COLLECTION · BIBLIOGRAPHY

Interruptions in Human-Computer Interaction

interruptions.net · a long-running online bibliography

A large list of research papers on interruptions, organized for browsing, with many of them available to read as PDFs. Several of the papers cited on this page are hosted there.

Why it is cited: as the place to keep reading, and as the host of several of the free copies linked above.

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→ interruptions.net
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C HOW IT WORKS (DETAILS) ✓

Prior know-how is assumed here, and the items become more "technical" toward the bottom.

The two lags, step by step

The picture at the top of the page is the standard way this research divides one interruption. Work on the main task runs until an **alert**: a phone rings, a message arrives. The **interruption lag** runs from the alert to the start of the interruption proper. Then comes the interruption itself, and after it the **resumption lag**, from the end of the interruption to the first action back on the main task[4].

The interruption lag is where preparation happens. In Trafton and colleagues' experiment, participants with an eight-second warning reported rehearsing where they were, and resumed faster than those interrupted with no warning; the second group closed some of the gap with practice[4]. The resumption lag is the measurable trace of how well that preparation, or the environment, holds one's place.

Reading "23 minutes" correctly

Mark, González and Harris observed information workers in the field and coded their activity at two levels: *events*, the use of one device or document or one conversation, and *working spheres*, the larger projects those events served[1]. Interrupted working spheres were resumed the same day about 82% of the time, after an average of 23 minutes 15 seconds, usually with about two other working spheres in between[2].

So the number is a *time to return*, measured across the day, not a *time to refocus*, measured from the moment of return. Quoted as the second, it overstates the cost of a single interruption; quoted as the first, it describes how fragmented the working day is. Both are worth knowing, and they are not the same thing.

What the laboratory adds, and where it disagrees

The field study suggested that interruptions on the same subject help[2]. The follow-up experiment found no difference by subject, and found interrupted people faster, at a cost in stress, frustration, time pressure and effort[3]. The two are not in direct conflict: one describes what workers reported over real days, the other what happened under controlled conditions in

one sitting. The safe reading is that the cost of an interruption does not always appear as lost minutes; it can appear as strain.

Timing is the other strong laboratory result. Adamczyk and Bailey placed interruptions at moments chosen from models of each task's structure, and found that the moment changed people's emotional state and their view of the interrupter[6].

Attention residue

Leroy's experiments asked people to switch from one task to another, finished or unfinished, and measured how much the first task stayed on their minds and how well they did the second[5]. Unfinished first tasks left more residue and worse performance on the next. In the language of the lags above, residue is part of why the resumption lag is not the whole cost: a task can be resumed promptly and still be done with less than full attention.

Two older models that still apply

Information foraging treats a person looking for information as a forager deciding where to look next and when to leave, guided by the scent of the cues in front of them[7]. A trip taken on weak scent and abandoned on arrival is a pure cost, which is why a preview that strengthens the scent before leaving saves a whole round trip.

The **keystroke-level model** estimates routine task times from standard operators, among them about 0.4 seconds to home the hand between keyboard and mouse[8]. It is the smallest unit of switching cost there is, and the one most often multiplied.

How this site handles its own sources

This page is built in the GYR-O format: plain answers on the left, every source previewed in the middle with the link last, and the detail here. Each numeral opens its card, and a Back arrow on the card returns to the numeral you came from, which is the resumption cue the research recommends, applied to reading. Whether each source is still reachable is checked by hand and recorded in `links.json`.

No analytics, no trackers, no cookies. This page sets nothing and stores nothing for its readers.

GYR-O = **G**reen · **Y**ellow · **R**ed · and sometimes **O**range

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